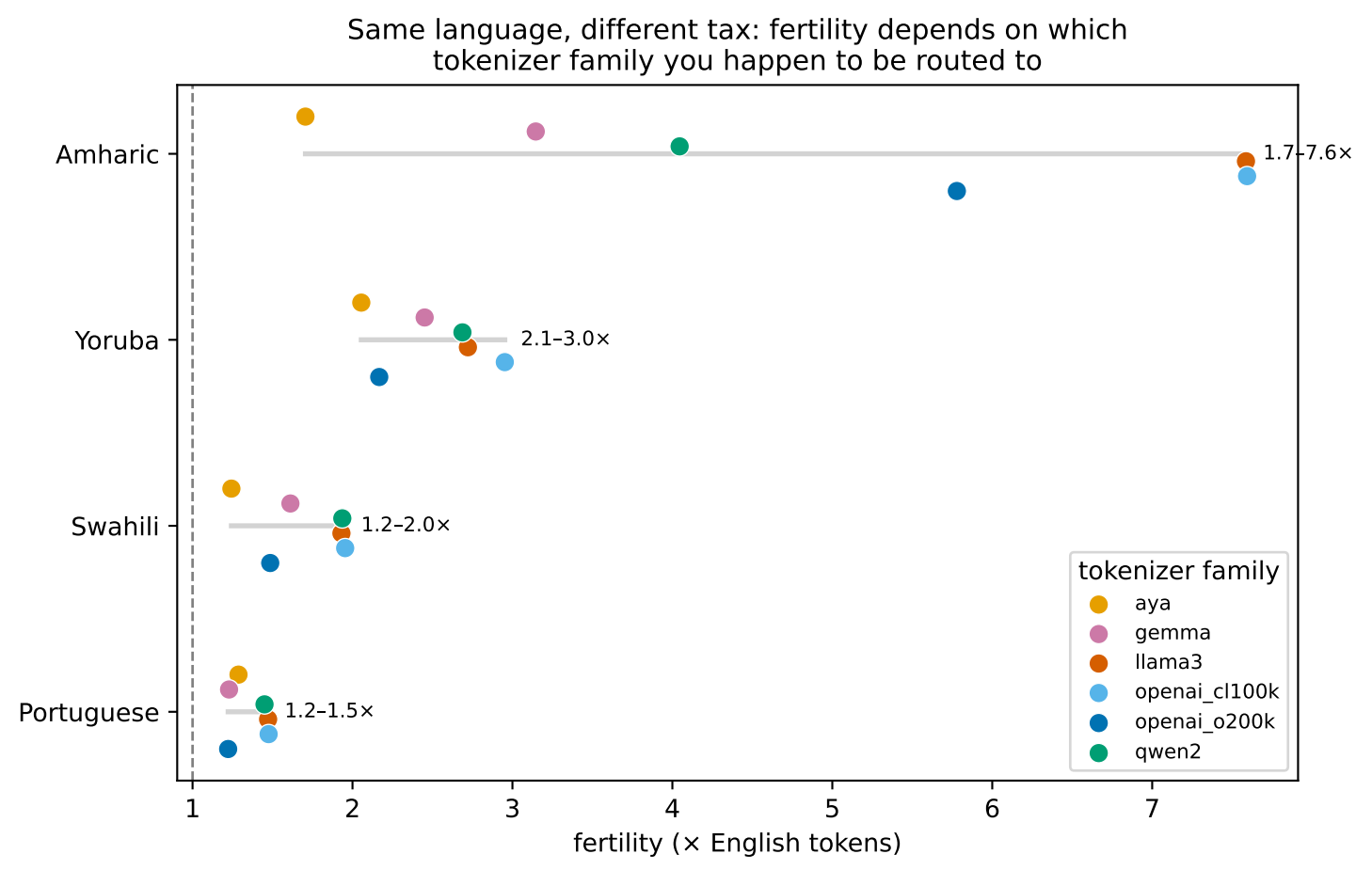
Same language, different tax: fertility depends on which tokenizer family you happen to be routed to
Amharic
Yoruba
Swahili
Portuguese
1.7–7.6×
2.1–3.0×
1.2–2.0×
1.2–1.5×
tokenizer family
aya
gemma
llama3
openai_cl100k
openai_o200k
qwen2
1
2
3
4
5
6
7
fertility (× English tokens)
FLORES-200 devtest · source Muennighoff/flores200 · measured 2026-07-06